



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech.(EVT)(CBCS) Degree Examinations
January 2023

22EVT122 – MODELING AND ANALYSIS OF ELECTRICAL MACHINES

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive the Mathematical model of separately excited DC motor and hence obtain the Transfer Function.	10	L*4	2	3,4,5
	b) Explain the working principles of synchronous machine without damper windings and hence obtain steady state equivalent circuit.	10	L2	1	3,4,5
2.	a) Derive the Mathematical model for a DC series motor.	10	L4	2	3,4,5
	b) Derive an expression for torque developed in a coil with saliency in a magnetic field. Assume the coil is supplied with voltage.	10	L4	1	3,4,5
3.	a) Why dynamic modeling of machine is necessary? Explain the steps involved in developing a model.	10	L2	1	3,4,5
	b) Derive the Two-phase power equivalent from three phase induction machine based on power equivalence principle.	10	L4	3	3,4,5
Unit – II					
4.	a) Derive an expression for torque developed in an induction motor in terms of rotor and stator d and q axes currents.	10	L4	3	3,4,5
	b) An induction motor has the following parameters: 2000hp, 230V, 3 phase, star connected, 4 pole, 60Hz, full load slip=0.03746. $R_s=0.02\Omega$; $R_r=0.12\Omega$; $X_m=50\Omega$; $X_{ls}=X_{lr}=0.32\Omega$ Find the electromagnetic torque developed by the machine when it is running at 1400 rpm using synchronous reference frames model & find the synchronous reference currents.	10	L4	3	3,4,5
5.	a) Derive an expression for normalized torque in dq modeling of induction motor.	10	L4	3	3,4,5
	b) With the help of per phase diagram, perform the per phase analysis of three phase connection of transformers for (i) Y-Y connection (ii) Δ - Δ connection	10	L4	4	3,4,5

P.T.O.

6. a) Consider the figure shown in FigQ6(a). Buses 1 and 2 are connected through parallel transformer T_1 with $X_1=j0.2$ and T_2 with $X_2=j0.4$. Bus 2 is a load bus supplying a current $I_{load}=1.05\angle-45^\circ$. $V_2=1\angle0^\circ$. Find the complex power transmitted to the load through each transformer if T_2 is a regulating transformer providing a 5% voltage magnitude boost. ($\bar{a}=1.05$)

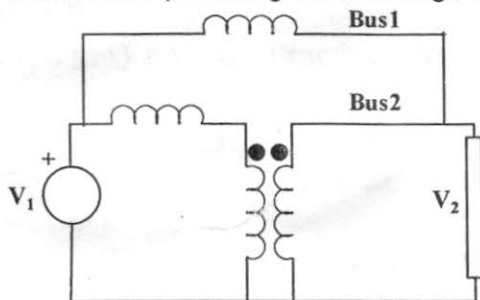


Fig Q6(a)

- b) What is meant by per unit normalization in power system? Derive the per phase equivalent circuit of transformer Y-Y connection.

10 L4 4 3,4,5

10 L4 4 3,4,5

Unit – III

7. a) Derive the Torque equations in terms of 3 phase synchronous machine variables.
 b) Write about the significance of rotor angle and angle between rotors of synchronous machine.
8. a) Derive the stator voltage equations of a synchronous machine in arbitrary reference frame.
 b) Derive the Torque equations of synchronous machine in per unit system.

10 L4 5 3,4,5

10 L4 5 3,4,5

10 L4 5 3,4,5

10 L4 5 3,4,5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
